

- Q-1 Reduce using K-Map: $f = \sum_m(2, 3, 6, 7, 8, 10, 11, 13, 14)$ [Ans. $f = A'C + B'C + CD' + AB'D' + ABC'D$]
- Q-2 Reduce using K-Map: $f = \sum_m(0, 1, 2, 3, 5, 7, 8, 9, 10, 12, 13)$ [Ans. $f = AC' + A'D + B'D'$]
- Q-3 Reduce using K-Map: $f = \sum_m(0, 2, 4, 6, 7, 8, 10, 12, 13, 15)$ [Ans. $f = A'D' + B'D' + ABC' + BCD$]
- Q-4 Reduce using K-Map: $f = \sum_m(0, 1, 2, 3, 6, 7, 13, 15)$ [Ans. $f = A'B' + A'C + ABD$]
- Q-5 Reduce using K-Map: $f = \sum_m(2, 3, 6, 7, 10, 11, 12)$ [Ans. $f = A'C + B'C + ABC'D'$]
- Q-6 Reduce using K-Map: $f = \sum_m(0, 1, 2, 3, 4, 5, 6, 9, 12, 13, 14)$ [Ans. $f = A'B' + C'D + BD'$]
- Q-7 Reduce using K-Map: $f = \sum_m(2, 3, 5, 7, 9, 11, 12, 13, 14, 15)$ [Ans. $f = AB + AD + BD + A'B'C$]
- Q-8 Reduce using K-Map: $f = \sum_m(5, 6, 7, 9, 10, 11, 13, 14, 15)$ [Ans. $f = BD + AD + BC + AC$]
- Q-9 Reduce using K-Map: $f = \sum_m(0, 1, 2, 3, 4, 6, 8, 9, 10, 11)$ [Ans. $f = B' + A'D'$]
- Q-10 Obtain the simplified expression using K-Map:
 $F = ABD + A'C'D' + A'B + A'CD' + AB'D$ [Ans. $f = A'D' + BD + AD$]